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Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No.: 5

Title: Building Sales Performance Dashboard using Power BI

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Experiment No.: 5

Building Sales Performance Dashboard using Power BI

Problem Statement

Nowadays, organizations generate a large amount of sales data every day. Analyzing this data manually is difficult and time-consuming. In this experiment, a Walmart sales dataset was used to create an interactive dashboard in Microsoft Power BI. The objective was to analyze sales, profit, quantity sold, and regional performance so that business information could be understood more easily through visual reports.

Methodology / Steps Followed

Step 1: Data Collection

The dataset contained information such as

- Order Date
- State, Category
- Product Name Sales
- Quantity
- Profit
- Ship Date.

Step 2: Data Import

- The dataset was imported into Power BI Desktop using the **Get Data** option. After loading the data, all the columns were checked to ensure they were imported correctly.

Step 3: Data Cleaning and Transformation

- Checked the dataset for missing values.
- Removed duplicate records wherever necessary.
- Verified and corrected data types such as Date, Text, and Decimal Number.
- Renamed columns for better readability.

Created calculated columns:

- Profit Status
- Shipping Days
- Walmart Sales Category (Low, Medium, High)

Step 4: Data Modeling

- Total Sales
- Total Profit
- Total Orders
- Total Quantity Sold
- Average Sales
- Average Profit
- Average Shipping Days

Step 5: Data Modeling

- Created KPI Cards for Average Sales, Average Profit, and Total Quantity Sold.
- Created Pie Chart for Profit by Category.
- Created Bar Chart for Sales by Category.
- Created Treemap for State-wise Sales.
- Created Line Chart for Monthly Sales Trend.
- Added Table Visual for detailed sales information.
- Added slicers for Category, State, and Year.
- Designed an interactive sales performance dashboard.

Step 6: DAX Measures and Calculated Columns

The following DAX measures and calculated columns were created to analyze the Walmart sales data and generate key business insights.

An interactive Power BI dashboard was designed using different visualization techniques.

- Measures used are:.

```
Total Orders = SUM(Walmart[Order ID])
```

```
Total Profit = SUM(Walmart[Profit])
```

```
Total Quantity Sold = SUM(Walmart[Quantity])
```

```
Total Sales = SUM(Walmart[Sales])
```

- Creating a new column “Profit Status”:

```

Profit Status =
IF(
    'Walmart'[Profit] > 0,
    "Profit",
    "Loss"
)

```

Profit Status
Profit
Profit
Profit
Profit
Profit
Loss

- Creating a new column “Wallmart Sales”:

```

Wallmart Sales = SWITCH(
    TRUE(),
    Walmart[Sales]>=1000,"HIGH",
    Walmart[Sales]>=500,"MEDIUM",
    "LOW"
)

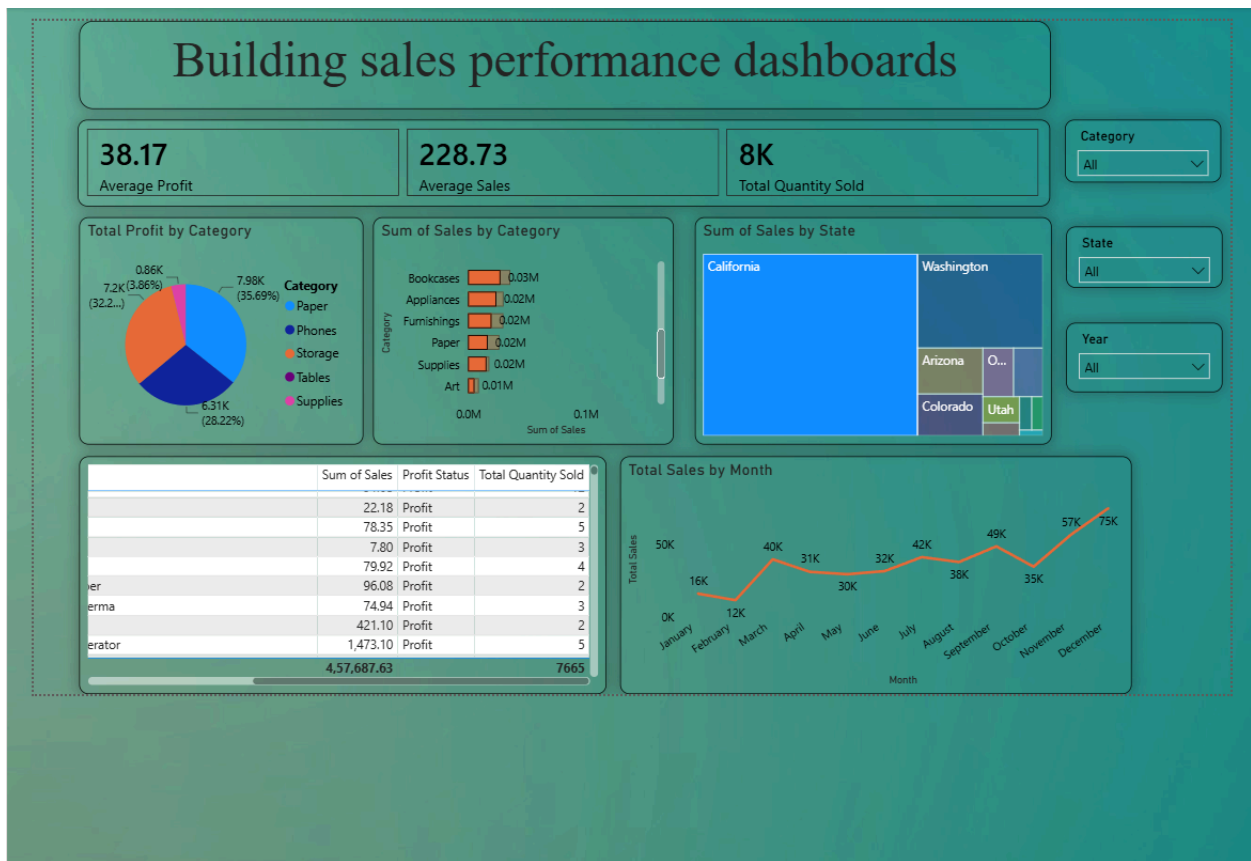
```

Walmart Sales
LOW
LOW
LOW
LOW
LOW
LOW
LOW
MEDIUM

Step 7: Dashboard Development

An interactive Power BI dashboard was designed using different visualization techniques.

Output



The final Power BI dashboard provides:

- Total Sales Performance
- Customer Analysis
- Product-wise Sales Analysis
- Regional Sales Performance
- Monthly Sales Trends
- Interactive Filters and Slicers

The dashboard enables management to monitor business performance and make data-driven decisions effectively.

Conclusion

This experiment successfully demonstrated the use of Microsoft Power BI for business data analysis and visualization. The Walmart sales dataset was imported, cleaned, transformed, and analyzed using DAX measures and calculated columns. Various charts, KPI cards, and slicers were used to build an interactive dashboard that presents business information in a simple and meaningful way.

The dashboard helps users monitor sales performance, compare profit across categories, analyze regional sales, and observe monthly trends. Overall, the experiment improved the understanding of data visualization techniques and business intelligence concepts using Power BI.

Learning Outcomes

Through this experiment, I learned:

- How to import Excel datasets into Microsoft Power BI.
- How to clean and transform business data using Power Query.
- How to verify and modify data types.
- How to create calculated columns.
- How to develop DAX measures for business analysis.
- How to build KPI cards for important business metrics.
- How to create Pie Charts, Bar Charts, Treemaps, Line Charts, and Table visuals.
- How to use slicers for interactive dashboard filtering.